

# JIWON JUNG

(she/her/hers)

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## RESEARCH INTERESTS

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**Stochastic modeling and machine learning for time series and event-driven data**, with applications to high-frequency financial markets, risk analytics, and industrial AI systems.

## EDUCATION

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**Ph.D. in Statistics**, Purdue University 2021–2024

Thesis: Dynamics of Modern Financial Markets: Data-Driven Approaches

(Co-)Advisor: Prof. Kiseop Lee and Prof. Mengyi Xu

**M.S. in Statistics**, Seoul National University 2017–2019

Thesis: Statistically Principled Crowdsourcing Method for Sports Highlight Selection

Advisor: Prof. Joong-ho Won

**B.S. in Statistics and B.A. in Economics**, Seoul National University 2013–2017

## PROFESSIONAL EXPERIENCE

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### Academic Appointments

University of Washington, Applied Mathematics: Pearson Fellow/Acting Instructor Sep. 2025–Present  
Teaching and research appointment in the Computational Finance & Risk Management (CFRM) program

Purdue University, Statistics: Visiting Assistant Professor Jan. 2025–Aug. 2025

Taught graduate and undergraduate statistics courses, including online and Indianapolis sections.

### Research & Industry Experience

VivifyAI: Smart Factory Data Analyst (remote) 2022–2024

Asan Medical Center: Cancer Center Research Assistant (Seoul, Korea) 2019–2020

LG CNS: Smart Factory Winter Intern (Seoul, Korea) 2018

## SELECTED AWARDS & HONORS

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I.W. Burr Award for excellence in dissertation research and results, Purdue University 2025

Outstanding Teaching Award for Teaching Assistant, Purdue University 2024

Recognition Award for Efficiency Improvements, Purdue University 2024

Emily and Paul Kidwell Graduate Student Excellence Award, Purdue University 2022

## PUBLICATIONS

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### Manuscripts Under Review

1. **Jung, J.** and Leung, T. (2026). Modeling the Interactions Between Zero-Day Options and Underlying Markets Using Joint Limit Order Book Data. Manuscript under review.

### Peer-Reviewed Publications

1. **Jung, J.**, Lee, K., and Xu, M. (2026). Modeling Multistate Health Transitions with a Most-Recent-Event Hawkes Process. *North American Actuarial Journal*, 30 (1), 157–178.
2. **Jung, J.** and Lee, K. (2025). Attention-based reading, highlighting, and forecasting of the limit order book. *Quantitative Finance*, 25 (7), 1015–1027. [**Feature Article**]

3. **Jung, J.**, Lee, K., and Leung, T. (2025). Threshold Overnight Comovement Analysis of Intraday and Overnight Returns. *Investment Analysts Journal*, 55 (2), 109–129.
4. Ho, D. J., Chui, M. H., Vanderbilt, C. M., **Jung, J.**, Robson, M. E., Park, C. S., and Fuchs, T. J. (2023). Deep Interactive Learning-Based Ovarian Cancer Segmentation of H&E-Stained Whole Slide Images to Study Morphological Patterns of BRCA Mutation. *Journal of Pathology Informatics*, 14, 100160.
5. **Jung, J.**, Ha, S., Son, W., Lee, J., and Won, J. H. (2022). SportLight: Statistically Principled Crowdsourcing Method for Sports Highlight Selection. *Journal of the Korean Statistical Society*, 51 (1), 127-148.
6. Shin, S. J., You, S. C., Jeon, H., **Jung, J. W.**, An, M. H., Park, R. W., and Roh, J. (2021). Style Transfer Strategy for Developing a Generalizable Deep Learning Application in Digital Pathology. *Computer Methods and Programs in Biomedicine*, 198, 105815.
7. Kim, S. W., Roh, J., **Jung, J.**, Pak, H. K., Lee, A. N., Park, Y. S., and Park, C. S. (2020). Immune Checkpoint Molecule V-set Ig Domain-Containing 4 (VSIG4) Expression is Associated with Poor Prognosis in Advanced Gastric Cancer Patients. *The Journal of Immunology*, 204, 243.4-243.4
8. Roh, J., **Jung, J.**, Lee, Y., Kim, S. W., Pak, H. K., Lee, A., and Park, C. S. (2020). Risk Stratification Using Multivariable Fractional Polynomials in Diffuse Large B-Cell Lymphoma. *Frontiers in oncology*, 10, 329.

## CONFERENCE PRESENTATIONS

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1. **Jung, J.** and Leung, T. (2026). Modeling the Interactions Between Zero-Day Options and Underlying Markets Using Joint Limit Order Book Data. Accepted for presentation, *Job Market Showcase Talk at INFORMS Annual Meeting 2026*, San Francisco, CA, U.S.
2. **Jung, J.**, Lee, K. (2024). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Invited talk at Joint Statistical Meetings 2024*, Portland, OR, U.S.
3. **Jung, J.**, Lee, K., and Xu, M. (2024). Modeling Multi-state Health Transitions with a Self Exciting Process. *Invited talk at American Mathematical Society (AMS) Sectional Meeting*, UMW, Milwaukee, WI, U.S.
4. **Jung, J.**, Lee, K., and Xu, M. (2023). Modeling Multi-state Health Transitions with Hawkes Processes. *Invited talk at INFORMS 2023*, Phoenix, AZ, U.S.
5. **Jung, J.**, Lee, K., and Xu, M. (2023). Modeling Multi-state Health Transitions with Hawkes Processes. *CEPAR International Conference*, UNSW, Sydney, Australia
6. **Jung, J.** and Lee, K. (2023). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Invited talk at SIAM Financial Mathematics and Engineering 2023*, DoubleTree by Hilton Philadelphia Center City, Philadelphia, PA, U.S.
7. **Jung, J.**, Leung, T., and Lee, K. (2023). A Lead-lag Analysis of Intraday and Overnight Returns. *Invited talk at American Mathematical Society (AMS) Sectional Meeting*, Georgia Institute of Technology, Atlanta, GA, U.S.
8. **Jung, J.**, Lee, K., and Xu, M. (2022). Modeling Functional Disability with Hawkes Process. *Actuarial Research Conference*, Urbana, IL.
9. **Jung, J.**, Roh, J., and Park, C. S. (2021). Abstract PO-079: Fused LASSO application for gastric cancer image segmentation. *Clinical Cancer Research*, 27, PO-079. *American Association of Cancer Research*, virtual.

## TEACHING EXPERIENCE

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### Instructor

Applied Mathematics Dept., University of Washington  
 — CFRM 405: Mathematical Methods for Quantitative Finance Fall 2025; Fall 2026 scheduled  
 — CFRM 521: Machine Learning for Finance Spring 2026  
 — CFRM 425: R Programming for Quantitative Finance Winter 2026  
 — CFRM 590: Multiscale Financial Signal Processing (co-instructor: Prof. Tim Leung) Fall 2025  
 Statistics Dept., Purdue University  
 — STAT 517: Statistical Inference (online) Summer 2025  
 — STAT 301: Elementary Statistical Methods (incl. Indianapolis section, S25) Spring 2023–Spring 2025

### Teaching Assistant

Statistics Dept., Purdue University  
 — STAT 303: Probability & Statistics for Business (led recitations) Fall 2021–Spring 2022  
 — STAT 511: Statistical Methods Spring 2021  
 — STAT 512: Applied Regression Analysis Spring 2021  
 College of Liberal Studies Dept., Seoul National University  
 — Selected Topics Seminar 2: Information Theory Fall 2018  
 — Selected Topics Seminar 1: Knowledge Spring 2017  
 Statistics Dept., Seoul National University  
 — Statistics Fall 2017  
 — Science Camp for High school Students in College of Natural Science Summer 2017  
 — Big Data Special Course using R Jan. 2016

## PROFESSIONAL SERVICE AND DEVELOPMENT

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### Professional Service

**Ad hoc reviewer:** *ASTIN Bulletin*; *European Journal of Finance*; *Studies in Economics and Finance*; 2024–2025  
*Journal of Data Science*

**Co-organizer, invited session:** “Data-Driven Methods in Financial Markets,” JSM 2024

### Mentoring and Student Service

Advised a three-student team for the COMAP Mathematical Contest in Modeling 2026  
 Mentored Purdue Undergraduate Research Conference participant 2025  
 Served as judge for undergraduate research presentations 2025

### Selected Travel Grants

George Casella Travel Award and ASA Travel Fund for JSM 2024  
 CEPAR Travel Grant 2023  
 SIAM Travel Grant 2023  
 Graduate Women in Science Program, Purdue University 2022

## TECHNICAL SKILLS & LANGUAGES

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Programming languages: Python, R (proficient); MATLAB, Julia, C/C++, Java (intermediate)  
 Statistical analysis tools: Excel, SPSS (proficient); SAS (intermediate)  
 Languages: English (fluent); Korean (native)